Experiment 6 : Set Up NGINX Load Balancing

<u>Step 1: Create Simple Node.js App</u>

1. Create a Project

mkdir load-balance-demo

cd load-balance-demo

npm init -y

npm install express

2. Create server.js

```
const express = require("express");
const app = express();

const PORT = process.env.PORT || 3000;

app.get("/", (req, res) => {
res.send(`Response from Server running on port ${PORT}`);
});

app.listen(PORT, () => {
console.log(`Server running on port ${PORT}`);
});
```

3. Run the application

node server.js

4. Create a Folder Load balancing

- Add node.js project

  change index.js file (server – 1)

  port : 3000

- Add node.js project

  change index.js file (server – 2)

  port : 3001

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Step 2: Run Multiple Servers

Open 2 terminals:

Terminal 1:

At PORT=3000
node server.js

Terminal 2:

at PORT=3001
node server.js

So we have now:

http://localhost:3000

http://localhost:3001

Step 3. Configure NGINX for Load Balancing (Open nginx.conf and Modify http block)

```
http {
   upstream myapp {
      server localhost:3000;
      server localhost:3001;
   }
   server {
      listen 8080;
      server_name localhost;
      location / {
         proxy_pass http://myapp;
      }
   }
}
```

Step 4: Restart NGINX

nginx -s stop

Nginx

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Now open:

http://localhost:8080

Refresh multiple times.

You can check:

Response from Server running on port 3000
Response from Server running on port 3001

Step 5: Make Server 1 Slow

Modify Server running on PORT 3000

```
app.get("/", (req, res) => {
  setTimeout(() => {
    res.send("Slow Server - Port 3000");
  }, 5000);
});
```

So Every request will wait 5 seconds before responding.

Keep Server 2 normal

```
app.get("/", (req, res) => {
  res.send("Fast Server - Port 3001");
});
```

Step 6. Do Testing

Open: http://localhost:8080

Now refresh quickly multiple times.

Even though server 3000 is slow,

NGINX still keeps sending every alternate request to it.

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Step 7: Now Change default algo of nginx(i.e. from round robin) to Least Connections and test again

Modify nginx.conf file:

```
upstream myapp {
   least_conn;
   server localhost:3000;
   server localhost:3001;
}
```

Restart NGINX: (use below commands)

```
nginx -s stop
nginx
```

Open: http://localhost:8080

Refresh quickly multiple times quickly.

When slow server (3000) is busy for 5 seconds:

It has an active connection.

So NGINX sees:

Server 3000 → 1 active connection

Server 3001 → 0 active connections

So it sends new requests to:

Fast Server (3001)